# ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)

**Lab Assignment 4:** Creation and Insertion in University Database.
Introduction to Databases (CSE 3151)
Programme: B.Tech. (CSE & CSIT) Semester: 6th

| Course Learning Outcome | Program Outcomes | Taxonomy Level |
|---|---|---|
| CO2: Model SQL queries and able to develop a database application as per user requirements using SQL and database connectivity. | PO1,PO2, PO5 | L2, L3 |

**Database Description:** A university database system maintains information about instructor, students, department, course, teaches, classroom, section, and takes. In this assignment, students will create tables using database constraints and insert sample records.

**Objective:** To understand SQL commands for table creation (CREATE) and data insertion (INSERT) using database constraints.

1. Create the following tables with their respective columns, data types and size:

| Table Name | Columns | CONSTRAINT DESCRIPTION |
|---|---|---|
| **DEPARTMENT** | DEPT_NAME | -PRIMARY KEY |
| | BUILDING | -NOT NULL |
| | BUDGET | -MUST BE > 0 |

| Table Name | Columns | CONSTRAINT DESCRIPTION |
|---|---|---|
| **INSTRUCTOR** | ID | -PRIMARY KEY<br>-LENGTH MUST BE 5 |
| | NAME | -NOT NULL |
| | DEPT_NAME | -FOREIGN KEY, REFERS TABLE DEPARTMENT |
| | SALARY | -MUST BE > 20000 |

| Table Name | Columns | CONSTRAINT DESCRIPTION |
|---|---|---|
| **STUDENT** | ID | -PRIMARY KEY<br>-LENGTH MUST BE 5 |
| | NAME | -NOT NULL |
| | DEPT_NAME | -FOREIGN KEY, REFERS TABLE DEPARTMENT |
| | TOT_CRED | -MUST BE > =0 |

| Table Name | Columns | CONSTRAINT DESCRIPTION |
|---|---|---|
| ADVISOR | I_ID | -PRIMARY KEY<br>-FOREIGN KEY, REFERS TABLE INSTRUCTOR |
| | S_ID | -PRIMARY KEY<br>-FOREIGN KEY, REFERS TABLE STUDENT |

| Table Name | Columns | CONSTRAINT DESCRIPTION |
|---|---|---|
| COURSE | COURSE_ ID | -PRIMARY KEY |
| | TITLE | -NOT NULL |
| | DEPT_NAME | -FOREIGN KEY, REFERS TABLE DEPARTMENT |
| | CREDIT | -MUST BE > 0 |

| Table Name | Columns | CONSTRAINT DESCRIPTION |
|---|---|---|
| CLASSROOM | ROOM_NO | -PRIMARY KEY |
| | BUILDING | -PRIMARY KEY |
| | CAPACITY | -MUST BE > =10 |

| Table Name | Columns | CONSTRAINT DESCRIPTION |
|---|---|---|
| SECTION | COURSE_ID | -PRIMARY KEY<br>-FOREIGN KEY, REFERS TABLE COURSE |
| | SEC_ID | -PRIMARY KEY<br>-NOT NULL |
| | SEMESTER | -PRIMARY KEY<br>-ANY OF SPRING, SUMMER, FALL, WINTER |
| | YEAR | -PRIMARY KEY<br>-MUST BE > =2000 AND <= 2026 |
| | ROOM_NO | -FOREIGN KEY, REFERS TABLE CLASSROOM |
| | BUILDING | -FOREIGN KEY, REFERS TABLE CLASSROOM |

| Table Name | Columns | CONSTRAINT DESCRIPTION |
|---|---|---|
| TEACHES | ID | -PRIMARY KEY<br>-FOREIGN KEY, REFERS TABLE INSTRUCTOR |
| | COURSE_ID | -PRIMARY KEY<br>-FOREIGN KEY, REFERS TABLE SECTION |
| | SEC_ID | -PRIMARY KEY<br>-FOREIGN KEY, REFERS TABLE SECTION |
| | SEMESTER | -PRIMARY KEY<br>-FOREIGN KEY, REFERS TABLE SECTION |
| | YEAR | -PRIMARY KEY<br>-FOREIGN KEY, REFERS TABLE SECTION |

| Table Name | Columns | CONSTRAINT DESCRIPTION |
|---|---|---|
| TAKES | S_ID | -PRIMARY KEY<br>-FOREIGN KEY, REFERS TABLE STUDENT |
| | COURSE_ID | -PRIMARY KEY<br>-FOREIGN KEY, REFERS TABLE SECTION |
| | SEC_ID | -PRIMARY KEY<br>-FOREIGN KEY, REFERS TABLE SECTION |
| | SEMESTER | -PRIMARY KEY<br>-FOREIGN KEY, REFERS TABLE SECTION |
| | YEAR | -PRIMARY KEY<br>-FOREIGN KEY, REFERS TABLE SECTION |
| | GRADE | |

2. Insert data into Department table with the given data.

| *dept_name* | *building* | *budget* |
|---|---|---|
| Biology | Watson | 90000 |
| Comp. Sci. | Taylor | 100000 |
| Elec. Eng. | Taylor | 85000 |
| Finance | Painter | 120000 |
| History | Painter | 50000 |
| Music | Packard | 80000 |
| Physics | Watson | 70000 |

3. Insert data into Instructor table with the given data.

| *ID* | *name* | *dept_name* | *salary* |
|---|---|---|---|
| 10101 | Srinivasan | Comp. Sci. | 65000 |
| 12121 | Wu | Finance | 90000 |
| 15151 | Mozart | Music | 40000 |
| 22222 | Einstein | Physics | 95000 |
| 32343 | El Said | History | 60000 |
| 33456 | Gold | Physics | 87000 |
| 45565 | Katz | Comp. Sci. | 75000 |
| 58583 | Califieri | History | 62000 |
| 76543 | Singh | Finance | 80000 |
| 76766 | Crick | Biology | 72000 |
| 83821 | Brandt | Comp. Sci. | 92000 |
| 98345 | Kim | Elec. Eng. | 80000 |

4. Insert data into Student table with the given data.

| *ID* | *name* | *dept_name* | *tot_cred* |
|---|---|---|---|
| 00128 | Zhang | Comp. Sci. | 102 |
| 12345 | Shankar | Comp. Sci. | 32 |
| 19991 | Brandt | History | 80 |
| 23121 | Chavez | Finance | 110 |
| 44553 | Peltier | Physics | 56 |
| 45678 | Levy | Physics | 46 |
| 54321 | Williams | Comp. Sci. | 54 |
| 55739 | Sanchez | Music | 38 |
| 70557 | Snow | Physics | 0 |
| 76543 | Brown | Comp. Sci. | 58 |
| 76653 | Aoi | Elec. Eng. | 60 |
| 98765 | Bourikas | Elec. Eng. | 98 |
| 98988 | Tanaka | Biology | 120 |

5. Insert data into Course table with the given data.

| *course_id* | *title* | *dept_name* | *credits* |
|---|---|---|---|
| BIO-101 | Intro. to Biology | Biology | 4 |
| BIO-301 | Genetics | Biology | 4 |
| BIO-399 | Computational Biology | Biology | 3 |
| CS-101 | Intro. to Computer Science | Comp. Sci. | 4 |
| CS-190 | Game Design | Comp. Sci. | 4 |
| CS-315 | Robotics | Comp. Sci. | 3 |
| CS-319 | Image Processing | Comp. Sci. | 3 |
| CS-347 | Database System Concepts | Comp. Sci. | 3 |
| EE-181 | Intro. to Digital Systems | Elec. Eng. | 3 |
| FIN-201 | Investment Banking | Finance | 3 |
| HIS-351 | World History | History | 3 |
| MU-199 | Music Video Production | Music | 3 |
| PHY-101 | Physical Principles | Physics | 4 |

5. Insert data into Section table with the given data.

| *course_id* | *sec_id* | *semester* | *year* | *building* | *room_number* |
|---|---|---|---|---|---|
| BIO-101 | 1 | Summer | 2009 | Painter | 514 |
| BIO-301 | 1 | Summer | 2010 | Painter | 514 |
| CS-101 | 1 | Fall | 2009 | Packard | 101 |
| CS-101 | 1 | Spring | 2010 | Packard | 101 |
| CS-190 | 1 | Spring | 2009 | Taylor | 3128 |
| CS-190 | 2 | Spring | 2009 | Taylor | 3128 |
| CS-315 | 1 | Spring | 2010 | Watson | 120 |
| CS-319 | 1 | Spring | 2010 | Watson | 100 |
| CS-319 | 2 | Spring | 2010 | Taylor | 3128 |
| CS-347 | 1 | Fall | 2009 | Taylor | 3128 |
| EE-181 | 1 | Spring | 2009 | Taylor | 3128 |
| FIN-201 | 1 | Spring | 2010 | Packard | 101 |
| HIS-351 | 1 | Spring | 2010 | Painter | 514 |
| MU-199 | 1 | Spring | 2010 | Packard | 101 |
| PHY-101 | 1 | Fall | 2009 | Watson | 100 |

6. Insert data into Classroom table with the given data.

| *building* | *room_number* | *capacity* |
|---|---|---|
| Packard | 101 | 500 |
| Painter | 514 | 10 |
| Taylor | 3128 | 70 |
| Watson | 100 | 30 |
| Watson | 120 | 50 |

7. Insert data into Teaches table with the given data.

| *ID* | *course_id* | *sec_id* | *semester* | *year* |
|---|---|---|---|---|
| 10101 | CS-101 | 1 | Fall | 2009 |
| 10101 | CS-315 | 1 | Spring | 2010 |
| 10101 | CS-347 | 1 | Fall | 2009 |
| 12121 | FIN-201 | 1 | Spring | 2010 |
| 15151 | MU-199 | 1 | Spring | 2010 |
| 22222 | PHY-101 | 1 | Fall | 2009 |
| 32343 | HIS-351 | 1 | Spring | 2010 |
| 45565 | CS-101 | 1 | Spring | 2010 |
| 45565 | CS-319 | 1 | Spring | 2010 |
| 76766 | BIO-101 | 1 | Summer | 2009 |
| 76766 | BIO-301 | 1 | Summer | 2010 |
| 83821 | CS-190 | 1 | Spring | 2009 |
| 83821 | CS-190 | 2 | Spring | 2009 |
| 83821 | CS-319 | 2 | Spring | 2010 |
| 98345 | EE-181 | 1 | Spring | 2009 |

8. Insert data into Takes table with the given data.

| *ID* | *course_id* | *sec_id* | *semester* | *year* | *grade* |
|---|---|---|---|---|---|
| 00128 | CS-101 | 1 | Fall | 2009 | A |
| 00128 | CS-347 | 1 | Fall | 2009 | A- |
| 12345 | CS-101 | 1 | Fall | 2009 | C |
| 12345 | CS-190 | 2 | Spring | 2009 | A |
| 12345 | CS-315 | 1 | Spring | 2010 | A |
| 12345 | CS-347 | 1 | Fall | 2009 | A |
| 19991 | HIS-351 | 1 | Spring | 2010 | B |
| 23121 | FIN-201 | 1 | Spring | 2010 | C+ |
| 44553 | PHY-101 | 1 | Fall | 2009 | B- |
| 45678 | CS-101 | 1 | Fall | 2009 | F |
| 45678 | CS-101 | 1 | Spring | 2010 | B+ |
| 45678 | CS-319 | 1 | Spring | 2010 | B |
| 54321 | CS-101 | 1 | Fall | 2009 | A- |
| 54321 | CS-190 | 2 | Spring | 2009 | B+ |

9. Insert data into Advisor table with the given data.

| *s_id* | *i_id* |
|---|---|
| 00128 | 45565 |
| 12345 | 10101 |
| 23121 | 76543 |
| 44553 | 22222 |
| 45678 | 22222 |
| 76543 | 45565 |
| 76653 | 98345 |
| 98765 | 98345 |
| 98988 | 76766 |